

Lights, Appliances, and Sunshine: A New HERS?

The HERS industry debates incorporating new end uses into a HERS score.

by Adam Gifford

For more than a decade now, a HERS rating has been calculated with an eye on the meat and potatoes of energy end uses. Only the energy use associated with heating, cooling, and the production of domestic hot water contributes to the making of a HERS score (see “The Current Construction of HERS”). It may be time to bring out the vegetables.

Discussions about expanding HERS to include other end uses have been ongoing for years, but recently the debate has become more pointed. Two new studies on the impact of incorporating lighting and appliance energy consumption, and even PV power generation, into HERS ratings have helped to ground the discussions.

Why the Debate?

From the beginning, HERS ratings were designed to rate only the permanent features of a home. “Rate the house, not the occupants” has been the often expressed industry mantra, implying that a home’s performance should not be distorted just because energy hogs live there. In keeping with this philosophy, two identical houses that have different thermostat setpoints, for example, should and will have the same HERS ratings.

So why consider changing how a rating is calculated? Because heating, cooling, and water heating are contributing to a shrinking portion of a home’s energy use, as homes become more and more efficient. A recent survey found that energy costs associated with lighting and appliances now typically account for nearly 50% of a home’s total energy expenditures (see Figure 1). In mild climates, this ratio can easily exceed 50%.



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Proponents of changing HERS argue that adding end uses to HERS ratings will just make these ratings more comprehensive. Altering HERS scores to include additional end uses will also promote the installation of a wider range of efficient products and systems.

On the other side of the debate, people who object to changing HERS ratings feel that the basis of the rating score should be permanent. They feel that a change to a known measuring stick would result in confusion for consumers. Others argue that lighting and appliance upgrades may compromise efforts to improve building shells and mechanical systems.

A Relative Impact

Including lighting, appliances, and PV systems in HERS ratings was first seriously discussed at the 2001 Residential Energy Services Network (RESNET) conference. Attendees there debated whether to include any new end uses, and if so, exactly which ones

would be most appropriate. Most people agreed that hard-wired light fixtures and refrigerators should be included. Other appliances, such as clothes washers and dishwashers, were more controversial. Refrigerators and sometimes dishwashers tend to be built into the design of the kitchen cabinetry, making them more permanent fixtures. Clothes washers and dryers are more apt to be moved from one house to the next when ownership changes hands.

In response to the discussions, two studies were conducted to look at the possible impacts of this change. One study, conducted by the Florida Solar Energy Center (FSEC), considered the impact of including hard-wired lights and refrigerators. The other study, performed jointly by the Joint Management Committee (JMC), a consortium of New England electric and gas utilities, Conservation Services Group (CSG), and Architectural Energy Corporation (AEC), considered the impact of including dishwashers. Although they used different resources, the two studies came up with

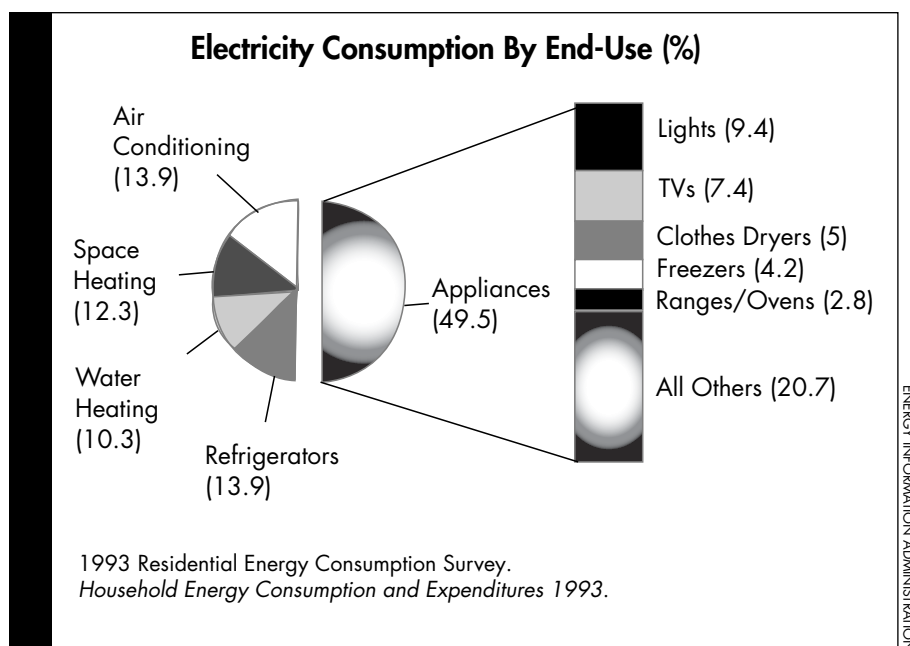


Figure 1. As they stand now, HERS scores miss out on substantial amounts of energy use.

similar baselines for lighting and appliance energy consumption, which suggests that those results are fairly reliable.

In the JMC/CSG/AEC study, which I worked on, two alternative HERS calculation methods were devised. One method employed the use of bonus points for altering HERS scores. This method only allowed scores to increase. The point score attributed to a particular level of efficiency was somewhat arbitrary, however. The other method rated PV generation and lighting and appliances energy use in the same way that the three primary end uses are rated—by comparing the efficiencies in the as-built home to baseline

efficiencies used in the reference home.

This second method, which we refer to as the expanded denominator approach, adds lighting and appliance baseline use to the reference home energy consumption, substantially increasing the overall coverage of energy use in the average North American home. However, it also means that for a new home to get the same score as it had originally earned, the builder has to improve the lighting and appliance efficiency to the same degree as the efficiencies for the rest of the building were improved. In discussions to date, the expanded denominator method has definitely emerged as the method of

choice, largely because it is more consistent with the current rating methodology than the first method.

In the case of PV systems, the HERS score was calculated by crediting the electric power generation against the overall energy usage for the end uses being analyzed in the home, after converting them all to like units (site Btu). (In the FSEC study, a factor was added to account for source generation.) This provides a HERS point credit for the production of renewable energy in the home, despite the fact that many homes use little or no electricity for such HERS-rated end uses as space and water heating.

A Rerater's Dream

In this same study, a sample of 119 homes in Massachusetts that were rated as part of the JMC's Energy Star Labeled Homes program were rerated using the alternative scoring methodologies described above (see Figure 2). Actual installed light fixtures and appliance data were collected as part of the final inspections on these units.

In general, the impact on HERS ratings of adding in new end uses is proportional to the contributions that these end uses make to the total energy budget of a home. The homes in our study are located in a substantial heating climate, and lighting and appliance energy use tends to account for only a small portion of total energy use. Overall, for these homes, the impact was fairly small. The average score went from 87.9 to 87.7—a difference of 0.17. Some scores remained unchanged. The

The Current Construction of HERS

HERS ratings, which were developed in the early 1990s with support from DOE, are used to gauge the relative energy efficiency of both new and existing homes on a standardized scale. A HERS rating is calculated by comparing the estimated energy use of an actual home to the energy use of a theoretical, reference home. The reference home is the same size and shape as the actual

home, but it is simulated using standards that would approximately meet the 1993 Model Energy Code.

The efficiency of the actual home is rated on a scale from 0 to 100. By definition, a home that uses the same amount of energy as the reference home will receive a score of 80 (or 4 stars). If a home uses 5% less energy than the reference home, one point is added to the HERS rating. Similarly, one point would be subtracted for a home that uses 5% more energy. A home that uses zero net

energy—100% less energy than the reference home—would receive a HERS rating of 100 (80 plus 20 points). A HERS rating of 86 (or 5 stars), which is approximately 30% more efficient than model codes, is required for a new home to earn an Energy Star label. Keep in mind, though, that these point scores only take into account the three end uses—heating, cooling, and water heating—and ignore other energy consumption in the modeled home.



The impact of the efficient appliances in this house was offset by the predominant use of incandescent lighting. As a result, the modified HERS score fell.

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most efficient homes generally decreased when the expanded denominator scoring method was used. On the other hand, these efficient lighting fixtures tended to raise the scores of homes that began with lower scores. The two trends pushed the ratings in different directions and tended to offset each other over the entire group.

For testing the impact of PV-generated power on the rating score, a certain amount of PV had to be assumed, as none of the rated buildings actually had PV installed. A system size of 5W/ft² of conditioned floor area was assumed for all buildings. Consequently, the overall impact on HERS ratings was much more consistent, with increases of approximately 1 point over the score with lighting and appliances included.

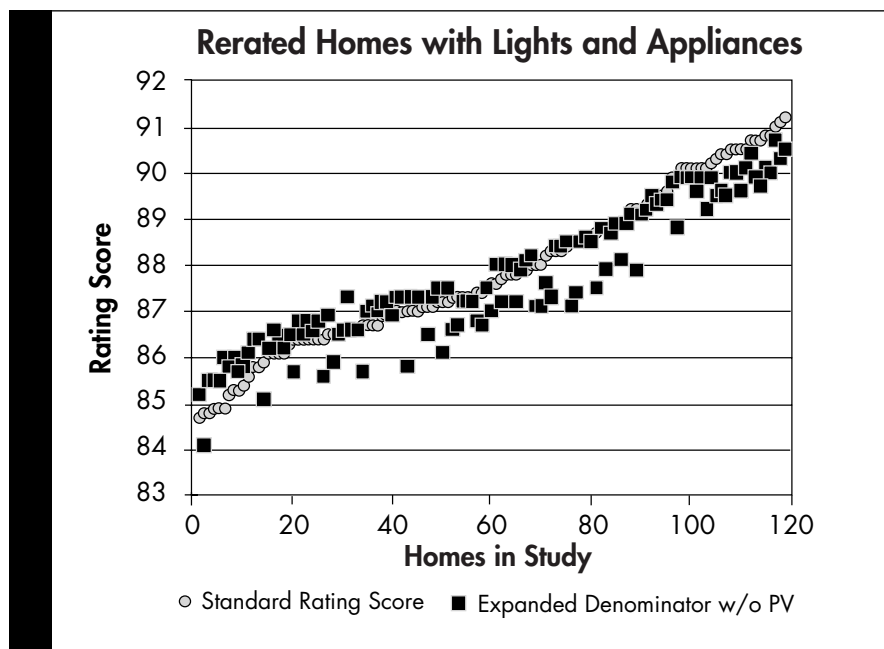
Mixed Reviews

The idea of changing HERS scores so that they give credit for efficient lighting, appliances, and PV systems has received mixed reviews. While some oppose revising the HERS score because of the confusion that may ensue, Dave Roberts of Architectural Energy Corporation argues that the basis of the rating score should not be locked in "if the change adds value to the system."

Before any changes are implemented, a broad industry consensus will need to develop on exactly how and when those changes should occur. Subsequently, the *National Home Energy Rating Technical Guidelines* will have to be modified. Look for this and other changes to occur as the HERS industry adapts to the changing profile of home energy use.



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Figure 2. Incorporating lighting and appliance energy use into reratings of these 119 Energy Star homes resulted in very small changes in their HERS scores.

largest point gain was +1.1. The largest point loss was -1.4.

The fact that the sample consisted exclusively of Energy Star homes influenced the study in two ways. On the one hand, the fact that the heating, cooling, and hot water systems were at a minimum 30% more efficient than the baseline appliances made it difficult for lighting and appliances to affect scores positively. In order for the scores to rise,

the newly included light fixtures and appliances in the rerated home would need to be more efficient than the space- and water-heating appliances. While the light fixtures and appliances found in this sample were generally more efficient than average, as a result of utility rebates provided to program participants, they were not more efficient than the heating appliances found in the most efficient homes. For this reason, the ratings for the

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